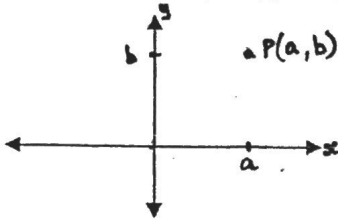


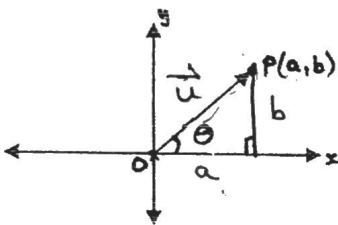
ALGEBRAIC VECTORS

Algebraic vectors allow the use of algebraic methods in the study of vectors.

Vectors in 2-dimensions (in a plane):



Point P in the plane is determined by the ordered pair which are its coordinates.



$\vec{OP}$ is the position vector of a point P from the origin to the point.

$$\vec{u} = \vec{OP} = [a, b]$$

square brackets

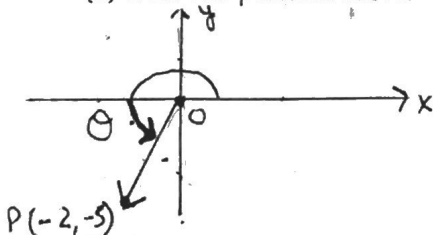
The ordered pair (a, b) is called an algebraic vector. The values of "a" and "b" are the "x" and "y" components of the vector.

The magnitude is $|\vec{u}| = \sqrt{a^2 + b^2}$.

The direction is $\theta = \tan^{-1}\left(\frac{b}{a}\right)$ with $0 \leq \theta \leq 360^\circ$ measured counter-clockwise from the positive x-axis to the line of the vector. The actual value depends on the quadrant in which $P(a, b)$ lies.

Example #1: Given point $P(-2, -5)$,

(a) draw the position vector



(c) determine the magnitude and direction

$$|\vec{OP}| = \sqrt{(-2)^2 + (-5)^2}$$

$$= \sqrt{29} \text{ exact}$$

$$\approx 5.39 \text{ units}$$

(b) express the position vector in ordered pair notation

$P(-2, -5)$ is in the 3rd quadrant.

$\therefore$ position vector is $\vec{OP} = [-2, -5]$

square brackets

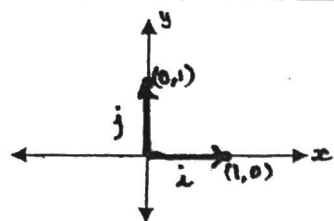
$$\tan(\theta) = \frac{-5}{-2}$$

$$\theta \approx 68.2^\circ \therefore 248.2^\circ \checkmark$$

$$\therefore \text{Bearing } 202^\circ \checkmark$$

$$S \therefore S22^\circ W \checkmark$$

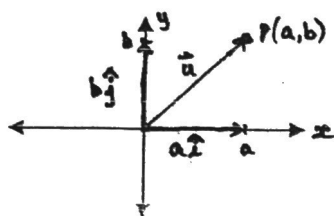
Notation using unit vectors:



carat

$\hat{i}$ is the unit vector along the positive x-axis and $\hat{j}$ is the unit vector along the positive y-axis.

Note: $\hat{i} = [1, 0]$ and $\hat{j} = [0, 1]$

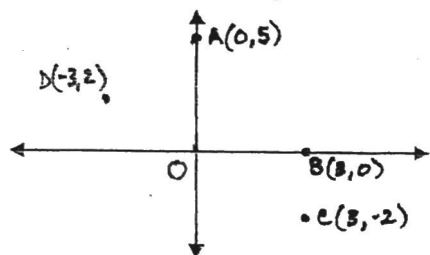


Ordered pair notation and unit notation are equivalent.

$\vec{u} = \overrightarrow{OP} = [a, b]$ or $\vec{u} = \overrightarrow{OP} = a\hat{i} + b\hat{j}$ thus, $[a, b] = a\hat{i} + b\hat{j}$

Vectors $a\hat{i}$ and $b\hat{j}$ are *vector components* of $\overrightarrow{OP}$ along $\hat{i}$ and $\hat{j}$ respectively. a and b are the *scalar components* of $\overrightarrow{OP}$.

Example #2: Express the position of each of the points shown as an ordered pair and in unit vector notation.



$$* \vec{OA} = [0, 5] = 0\hat{i} + 5\hat{j} = 5\hat{j}$$

$$* \vec{OB} = [3, 0] = 3\hat{i} + 0\hat{j} = 3\hat{i}$$

$$* \vec{OC} = [3, -2] = 3\hat{i} - 2\hat{j}$$

$$* \vec{OD} = [-3, 2] = -3\hat{i} + 2\hat{j}$$

Vector Addition and Subtraction in Component form:

If $\vec{u} = [u_1, u_2]$ and $\vec{v} = [v_1, v_2]$ then $\vec{u} + \vec{v} = [u_1 + v_1, u_2 + v_2]$ and $\vec{u} - \vec{v} = [u_1 - v_1, u_2 - v_2]$

Scalar Multiplication in Component form:

If $\vec{u} = [u_1, u_2]$ and $k \in \mathbb{R}$, then $k\vec{u} = [ku_1, ku_2]$ eg: $3\vec{u} = [3u_1, 3u_2]$
 $= 3[u_1, u_2]$

Two algebraic vectors are equal if and only if their respective Cartesian components are equal.

Example #3: If $\vec{u} = [2, -3]$ and $\vec{v} = [5, -3]$, find (a) $\vec{u} + \vec{v}$, (b) $-3\vec{u}$

$$= [7, -6] = [-6, 9]$$

$$= -3[2, -3]$$

$$c) 4\vec{u} - \vec{v} = [8, -12] - [5, -3]$$

$$= [3, -9]$$

$$= 3[1, -3]$$

Example #4: If $P_1 = (2, -3)$ and $P_2 = (-5, -2)$, find $\vec{P_1P_2}$

$$\vec{P_1P_2} = [-5 - 2, -2 - (-3)] = [-7, 1]$$

Example #5: Using vectors, show that the three points A(5, -1), B(-3, 4) and C(13, -6) are collinear.

* 3 points are collinear if $\vec{AB}$ and $\vec{BC}$ have the same or opposite direction.

$$\begin{cases} \vec{AB} = [-8, 5] \\ \vec{BC} = [16, -10] \end{cases} \therefore \vec{BC} = -2\vec{AB} \Rightarrow A, B \text{ and } C \text{ are collinear.}$$